

Computer Networking: Security (CSE 3752)

Assignment – 2

Q.1. Illustrate the importance of **race condition** in Cyber Security. Discuss, how it occurs in concurrent systems and also it can lead to incorrect or unpredictable results. Mention any two techniques used to prevent race conditions.

Q.2. A program crashes when excessive input is given, and attackers use it to execute malicious code. Identify and explain the vulnerability.

Q.3. State, how the SQLI (Structured Query Language Injection) a common possible cyber-attack works to cause harm during web browsing. Also narrate the mitigation process to prevent such type of attack.

Q.4. An employee account in a cloud system has excessive permissions and gets compromised, allowing attackers to access critical resources. Identify and explain the vulnerability.

Q.5. An application in a mobile device downloaded from an unknown source appears legitimate but secretly collects sensitive user information. Identify and explain the threat due to which the said vulnerability occurs. How users can avoid such threats.

Q.6. An employee at a marketing firm receives an email from what looks like a trusted vendor. The email contains a link to a 'New 2026 Budget Template.exe'. The employee clicks the link and runs the file. Immediately, a hidden script installs a keylogger to steal their passwords, but the file does not attempt to spread itself to other computers on the office Wi-Fi.

Based on the propagation method and the level of user interaction described, is this attack an example of a **Trojan horse** or a **Computer Worm**? Explain your reasoning by highlighting the specific behaviors in the scenario that distinguish the two.

Q.7. What is meant by DNS cache poisoning associated with DNS attack? Discuss briefly, how DNS sinkhole can be used to prevent DNS attack.

Q.8. Explain the following processes adopted by adversaries to launch an on-path attack in computer network.

(i) Session replay (ii) Message replay (iii) Credential replay (iv) Credential stuffing

Q.9. A system administrator detects multiple unknown administrator accounts being created after a low-privilege system was compromised. Specify the indicator of this attack. Justify, how does it help the attacker to maintain access?

Q.10. List and discuss briefly different types of segmentation techniques used in computer networking to reduce the impact of potential breach.

Q.11. Name the two key factors required to be executed to satisfy the goal of access control mechanism associated with security issue in computer networking. Also differentiate between a file System Access Control List (ACL) and a network ACL being a part of access control mechanism.

Q.12. A marketing company allows employees to browse the internet for research but needs to prevent access to known malware-hosting sites, gambling websites, and phishing sites. Should a whitelist or a blocklist be implemented on the company firewall to manage this? Justify your answer.

Q.13. State the importance of SIEM. A company notices multiple failed login attempts across different systems at unusual hours. Individually, these events seem harmless. How can SIEM help detect a potential attack?

Q.14. List and highlight the use of key elements for a multilayer strategy required to be implemented towards achieving effective network security.

Q.15. Briefly explain various methods of Infrastructure as Code (IaC) to develop a reliable, and efficient IT systems? Also explain what risks does Infrastructure as Code introduce?

Q.16. An attacker gains physical access to an IoT device and extracts sensitive data. Why is physical security important in IoT? Suggest preventive measures.

Q.17. With the help of suitable diagram, explain the function of each of the four Supervisory Control and Data Acquisition (SCADA) system levels.

Q.18. An organization hosts a public-facing web application and an internal employee database. The security team wants to allow external users to access the web application but strictly prevent direct access to internal systems. How the network zones (LAN, DMZ, WAN) should be structured, and where should each system be placed? Justify your design.

Q.19. Justify that data classification seems to be important in organizations for giving safeguards to various data types from security breach. Also mention the key classification types of data sets available in organizations.

Q.20. A hospital stores patient medical records, including diagnostic reports and personal details, in a cloud system. It needs to share selected data with external researchers while complying with strict privacy laws. Identify the **data type, classification**, and recommend **two security methods** to safely share the data. Justify your choice.

Q.21. A software company develops proprietary source code for a new application. It wants to prevent competitors and unauthorized users from understanding or stealing the code. Identify the **type and classification of data**, and recommend **two protection techniques**, explaining how each reduces risk

Q.22. Explain the following methods that are meant for securing data from possible breaches during communication over a network.

(i) Geographic restriction (ii) Hashing (iii) Tokenization (iv) Obfuscation

Q.23. A large organization experiences a malware attack that spreads rapidly across all systems due to a flat network architecture. Identify the missing security strategy, explain how it works, and describe **two benefits** in limiting attack impact.

Q.24. State the use of network load balancer that guarantees the availability of network to the host devices in a company even though the network carries high volume of traffic. Also differentiate the basic two load balancer configurations active/active and active/passive.

Q.25. A user logs into an online banking system. The load balancer is configured with **Affinity (sticky sessions)**.

- a) Explain how the load balancer will handle multiple requests from this user.
- b) Why is affinity important for applications like online banking?
- c) What problem may occur if affinity is not used?